FABRICATION AND COMPREHENSIVE CHARACTERIZATION OF BIOCHAR-ENHANCED PLA COMPOSITE 3D FILAMENT

Abstract

Advancements in fused filament fabrication (FFF), a form of additive manufacturing, have spurred interest in eco-friendly polymer composites. This study hypothesizes that incorporating biochar, a carbon-rich waste-derived filler, into polylactic acid (PLA) will enhance wear resistance, thermal conductivity, and mechanical properties. To test this hypothesis, PLA pellets were fabricated with 5 wt.% and 10 wt.% PLA–biochar using twin-screw extrusion and printed via FFF. Standard test specimens (one sample per test owing to material and cost constraints) were evaluated for mechanical, tribological, and thermal performance in accordance with ASTM standards. Tensile strength decreased from 53.33 MPa (pure PLA) to 12.83 MPa (5% biochar) but recovered slightly to 14.30 MPa at 10% biochar. Flexural strength followed a similar trend, decreasing from 81.33 MPa to 28.47 MPa (5%) and then increasing to 36.60 MPa (10%). Wear resistance improved significantly, with the wear rate dropping from 0.375 µm/s to 0.125 µm/s and the coefficient of friction falling from 0.375 to 0.095. Thermal conductivity rose from 0.1765 W/mK to 0.2414 W/mK at 10% biochar content. Scanning electron microscopic analysis revealed lowered material loss and improved surface integrity in biochar-filled samples. These results confirm that biochar enhances the functional properties of PLA composites, particularly in terms of wear resistance and thermal conductivity, supporting its use in sustainable, high-performance FFF applications.

Keywords: Fused filament fabrication (FFF), biochar–PLA filament, mechanical properties, wear resistance, thermal conductivity, twin-screw extrusion, scanning electron microscopy (SEM)

1 Introduction

Fused filament fabrication (FFF), a subset of additive manufacturing (AM), fabricates components layer by layer from digital models via the extrusion of thermoplastic filaments. Of particular relevance are recent developments in biomedical uses of FFF—such as the manufacture of 3D scaffolds for cell culture, surgical models, and patient-specific prostheses [35]. The cost-effectiveness, versatility in producing complex geometries, and minimal material waste of FFF have made it a preferred technique for prototyping and low-volume production [1]. Compared with conventional subtractive methods such as milling or turning, FFF enables more rapid design iteration and supports a wider variety of material systems, especially polymer composites.

Polylactic acid (PLA) is a biodegradable thermoplastic polymer derived from renewable feedstocks such as corn starch or sugarcane. PLA is attractive for use in FFF owing to its remarkable strength, ease of processing, and environmentally friendly characteristics. Nonetheless, its shortcomings—particularly brittleness, low thermal stability, and insufficient wear resistance—limit its application in functional or load-bearing applications [2].

Biochar, a carbon-rich material obtained via the pyrolysis of biomass under low-oxygen conditions, exhibits high porosity, thermal stability, and tunable conductivity. When incorporated into polymers, biochar can serve as a sustainable and low-cost reinforcement, augmenting wear resistance, thermal performance, and in some cases, electrical conductivity [3,4]. However, the effectiveness of biochar depends highly on uniform dispersion and interfacial adhesion with the polymer matrix, necessitating the use of compatibilizers or surface modification [5].

PLA has gained widespread attention owing to its biodegradability and renewability, with applications ranging from food packaging and disposable consumer products to biomedical uses such as sutures and scaffolds. Furthermore, PLA is the most commonly employed filament in FFF, accentuating its relevance in AM research [36,37].

Blending PLA with biochar offers a method to overcome its intrinsic weaknesses while maintaining its biodegradability. Biochar-reinforced PLA composites demonstrate improved stiffness, wear resistance, and thermal conductivity, thereby extending PLA's applicability to semistructural and functional engineering applications. In this study, PLA–biochar composite filaments containing 5 wt.% and 10 wt.% biochar were developed, and their mechanical, wear, and thermal performance were assessed against those of neat PLA.

PLA (procured from Terrasafe, Amazon, India) has a granule size of 3 mm and exhibits well-defined thermomechanical properties, characterized by a density of 1.24 g/cm³, low porosity, and moderate thermal stability, degrading at approximately 200°C. Its mechanical profile includes a tensile strength of 50–70 MPa, compressive strength of 50–100 MPa, and elastic modulus of 2–3 GPa, making it suitable for load-bearing applications in polymer-based composites. Biochar (supplied by Biochar Factory, Chennai, India) exhibits highly distinct characteristics, including a significantly lower bulk density (0.2–0.6 g/cm³) and a highly porous structure, which contribute to its large surface area and potential as a reinforcement or filler material [25,26]. Unlike PLA, biochar possesses excellent thermal stability, withstanding temperatures of up to 700–900°C, depending on the feedstock and pyrolysis conditions [27,28]. However, its mechanical strength is considerably lower, with compressive strength of only 1–5 MPa and very low tensile strength, rendering its raw form inappropriate for direct load-bearing applications [26,31]. In addition, biochar is generally electrically insulating but demonstrates semiconductive behavior if activated or graphitized [29,30]. These contrasting properties suggest that combining PLA with biochar would yield a balanced material system that leverages PLA's strength and processability along with biochar's high thermal stability, porosity, and sustainability [32].

Considerable advancements have been made in processing composite filaments for AM applications. While stereolithography offers fine resolution in ceramics, the challenges of post-processing shrinkage and low strength must be overcome [25]. Conversely, FFF accommodates a wider selection of thermoplastics, including PLA and acrylonitrile butadiene styrene (ABS) but suffers from issues such as binder dispersion and inconsistent flow. Formulation improvements, such as the PEG–POM–PMMA–SA binder system, have been shown to enhance extrusion quality in metallic composites [26]. Such advances provide a framework for optimizing bio-based fillers, such as biochar.

In this study, PLA–biochar composite filaments were developed using twin-screw extrusion and characterized, and standardized specimens were fabricated via FFF. Most previous studies have focused on mechanical testing. In contrast, this research provides a comprehensive multiproperty evaluation that encompasses mechanical, tribological, thermal, and morphological characteristics of 5% and 10% PLA–biochar composites. The need for this work arises from the escalating demand for sustainable, cost-effective reinforcements that can augment the functional performance of PLA without relying on expensive or nonbiodegradable additives. The working hypothesis is that biochar reinforcement can markedly enhance wear resistance and thermal stability, in addition to altering tensile and flexural strength. By elucidating how biochar content influences performance, this study offers novel insights into designing high-performance, eco-friendly polymer composites for engineering and industrial applications.

2 Methodology

To evaluate the effect of biochar reinforcement, neat PLA and PLA composites containing 5 wt.% and 10 wt.% biochar were fabricated and characterized. For each property (tensile, flexural, impact, thermal conductivity, and wear), one specimen was tested per composition. Although statistical averaging could not be performed due to resource allocations, the results were representative of the material's behavior.

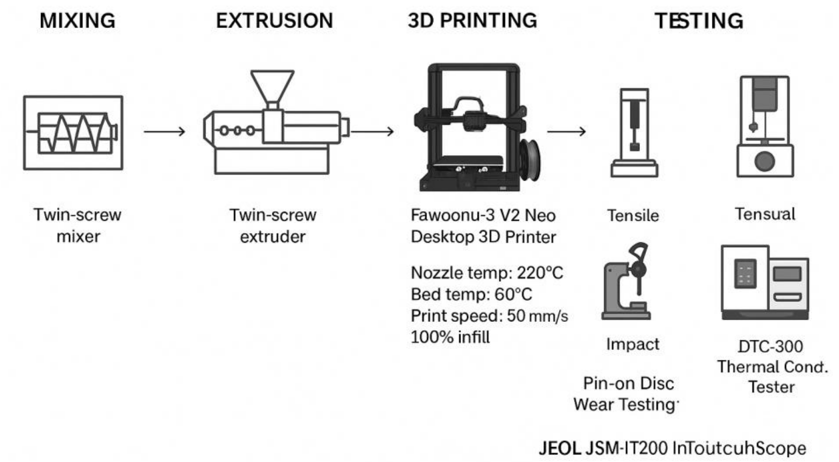


Fig. 1 Schematic of Fabrication and Testing Workflow

2.1 Twin-screw batch mixing and extrusion process

Biochar was incorporated into PLA at loading levels of 5 wt.% and 10 wt.%. These specific concentrations were selected based on earlier investigations establishing that moderate additions in this range improve the stiffness, tensile strength, and thermal stability of PLA composites. Furthermore, issues such as filler agglomeration and deterioration of melt flow properties, which negatively affect printability, are simultaneously minimized [25–27]. Prior to processing, both PLA pellets and biochar powder were oven-dried at 60°C for 4 h to prevent hydrolytic degradation during melt processing.

The dried constituents were premixed and compounded in a laboratory-scale corotating twin-screw batch mixer (Thermo Scientific Process 11, Germany) operated at 180°C and 60 rpm for 10 min to ensure uniform dispersion of the biochar within the polymer matrix. Subsequently, the compounded material was cooled to room temperature, pelletized, and extruded into filaments using the same corotating twin-screw extruder housed at the Central Institute of Plastics Engineering & Technology, India. The extrusion was performed at a barrel temperature of 170–200°C with a screw speed of 50 rpm, using a circular die with a diameter of 1.75 mm. The extrudates were immediately water-cooled, solidified, and spooled to obtain continuous filaments suitable for FFF 3D printing [6,7].

2.2 3D modeling and printing of the test specimens

The test specimens for tensile, impact, flexural, wear, and thermal conductivity tests were designed using CATIA V5, a widely adopted computer-aided design (CAD) platform in engineering applications [8]. To ensure standardization and repeatability, each specimen was modeled according to the respective ASTM standard, as follows:

- Wear test: ASTM G99-17 [9]
- Tensile test: ASTM D638 [10]
- Flexural test: ASTM D790 [11]
- Impact test: ASTM D256 [12]
- Thermal conductivity test: ASTM C518 [13]

The 3D CAD models are not included for brevity; however, they strictly followed the dimensional and geometric specifications outlined in the above standards. The PLA–biochar test specimens for tensile, wear, thermal conductivity, flexural, and impact tests were 3D printed using the Fawoonu-3 V2 Neo Desktop 3D Printer [14]. Various extrusion parameters were trialed to achieve optimal print quality and mechanical reliability. As biochar increases the melt viscosity, a nozzle temperature of 220°C was selected to achieve adequate flow and fusion of the PLA–biochar composite. The bed temperature was set at 60°C to minimize warping and improve first-layer adhesion. Furthermore, considering the particulate nature of biochar, a print speed of 50 mm/s was chosen to maintain dimensional accuracy and ensure consistent filament deposition. A 100% infill density was used to preserve the mechanical integrity of the test specimens. These parameter choices were guided by prior studies on PLA composites and adjusted based on initial printing trials for defect-free prints. [33, 34]

2.3 Tensile test

Tensile tests were performed using a tensometer (Model: H10K, Manufacturer: Mecmesin, India) at a testing speed of 5 mm/min to determine the tensile stress and strain of pure PLA, 5% PLA–biochar, and 10% PLA–biochar filaments. The applied force and specimen elongation were recorded to calculate tensile stress and strain, providing information on the effects of biochar on the material's strength and deformation behavior [10].

(a) (b)

Fig. 2 Tensile testing specimens (a) Before testing (b) After testing

2.4 Flexure test

Flexural tests were conducted on a tensometer at a testing speed of 2 mm/min to determine the bending behavior of pure PLA, 5% PLA–biochar, and 10% PLA–biochar filaments. The applied force and specimen deflection were recorded to understand the influence of biochar on the material's stiffness and flexural strength [11].

(a) (b)

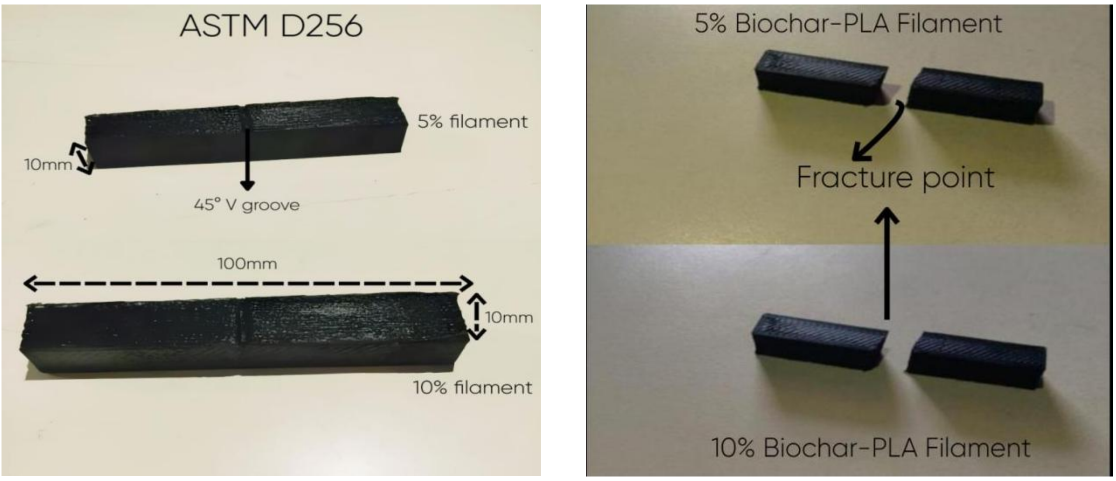
Fig. 3 Flexure testing specimens (a) Before testing (b) After testing

2.5 Impact test

Impact tests were performed on a Charpy impact tester (Model: R1, Manufacturer: ITW Test & Measurement, India) using the R1 scale to estimate the impact resistance of pure PLA, 5% PLA–biochar, and 10% PLA–biochar filaments. The energy absorbed during fracture was documented to determine the effect of biochar on the material's toughness [12].

The following formula was used to calculate impact strength:

$$\text{Impact strength} = \frac{\text{Impact energy}}{\text{Specimen width}}$$



(a) (b)

Fig. 4 Impact testing specimens (a) Before testing (b) After testing

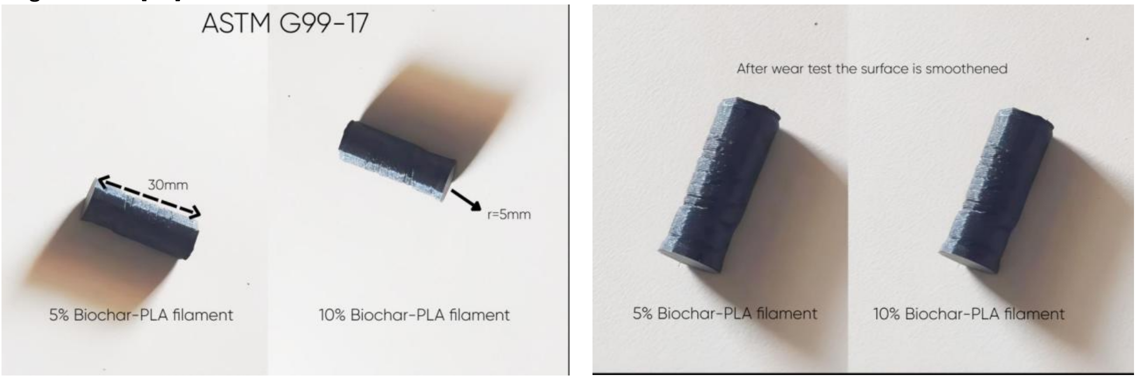
2.6 Wear test

Wear tests were performed using a pin-on-disk wear testing machine (Model: TR-20, Manufacturer: TriboTech Instruments, India) to assess the wear resistance of pure PLA, 5% PLA–biochar, and 10% PLA–biochar filaments. The parameters of material loss and frictional behavior were estimated to comprehend the effect of biochar on wear performance [10].

Based on similar studies and ASTM G99-17 guidelines [9], the following standardized test parameters were employed:

- Test environment: dry sliding condition (no lubrication)
- Load applied: 10 N
- Sliding speed: 1 m/s
- Total sliding distance: 1000 m
- Angular velocity of the disk: 20 rad/s
- Test duration: 16 min and 40 s (1000 m/1 m/s)

These values were selected owing to their common use in the tribological evaluation of thermoplastics and their composites, ensuring sufficient surface interaction without causing excessive wear or thermal degradation [33].



(a) (b)

Fig. 5 Wear testing specimens (a) Before testing (b) After testing

2.7 Thermal conductivity test

Thermal conductivity tests were conducted on a DTC-300 Thermal Conductivity Tester (Manufacturer: HeatFlow Instruments, India) to determine the heat transfer properties of pure PLA, 5% PLA–biochar, and 10% PLA–biochar filaments. The effect of the biochar content on the material's thermal performance was ascertained by recording and analyzing the thermal conductivity values [13].

(a) **(b)**

Fig. 6 Thermal conductivity testing specimens (a) Before testing (b) After testing

2.8 Scanning electron microscopy (SEM)

SEM was used to examine the surface of the PLA–biochar samples after wear testing. Images were acquired for both 5% and 10% biochar samples to determine the changes in the surface morphology. The study was conducted using a JEOL JSM-IT200 InTouchScope (Manufacturer: JEOL Ltd., India) at an accelerating voltage of 15 kV. Before imaging, the samples were coated with a thin layer of gold to make them conductive and obtain clear images. Different zoom levels were used to explore the surface features, including grooves, cracks, material smearing, and areas where particles were dislodged [22].

3 Results

Tensile, flexural, impact, wear, and thermal conductivity tests were performed on pure PLA, 5% PLA–biochar, and 10% PLA–biochar filaments. The performance of biochar-reinforced PLA was compared with that of pure PLA to assess the effects of biochar on the polymer's mechanical, wear, and thermal characteristics.

3.1 Tensile test

Tensile tests were conducted on pure PLA, 5% PLA–biochar, and 10% PLA–biochar filaments to evaluate their tensile properties. A gradually increasing load was applied using a tensometer until the specimen fractured. Tensile stress and strain were the key parameters measured, which helped determine the material's strength and deformation behavior. The results were analyzed to understand the influence of biochar on the mechanical performance of PLA. Tensile stress and strain were calculated using the measured load and elongation, as shown in Table 2.

Table 1 Tensile test results

Sample	Load (N)	Elongation (mm)	Tensile Stress (MPa)	Tensile Strain	Young's Modulus (MPa)
5%PLA–biochar	770	2.5	12.83	0.016	801.88
10% PLA–biochar	870	4.5	14.5	0.030	483.33
Pure PLA	3200	4.32	53.33	0.028	1904.64

The tensile strength of PLA decreased considerably with the incorporation of biochar, dropping from 53.33 MPa for pure PLA to 12.83 MPa and 14.5 MPa for 5% and 10% PLA–biochar, respectively. This reduction agrees with the findings of Zouari et al. [33], who noted tensile strength enhancement at 5 wt.% followed by reductions due to potential agglomeration. The tensile strength fell from 0.028 (pure PLA) to 0.016 at 5%, suggesting stiffening, but increased to 0.030 at 10%, indicating enhanced ductility at higher filler content. Similar trends have been reported by Yan et al. [34]. In their study, low biochar loadings improved ductility, although higher loadings led to stress-localized deformation. However, as the strain values were relatively low and the fractured surface was not analyzed, these findings should be considered preliminary and require further validation.

3.2 Flexure test

Flexural tests were conducted to assess the bending strength and deformation behavior of pure PLA, 5% PLA–biochar, and 10% PLA–biochar filaments. The test was performed using a tensometer, where the key parameters of applied load (N) and elongation (mm) were measured. Flexural stress and strain were then calculated using standard formulas. Young's modulus in bending, also known as the flexural modulus, quantifies the material's stiffness during flexural loading and is computed as the ratio of flexural stress to flexural strain within the linear elastic region of the stress–strain curve.

Table 2 Experimental data of flexure test

Sample	Load (N)	Elongation (mm)	Flexural Stress (MPa)	Flexural Strain	Young's Modulus (MPa)
5% PLA–biochar	70	17.1	28.47	0.0827	**344.18**
10% PLA–biochar	90	11.8	36.60	0.0571	**640.45**
Pure PLA	200	8.9	81.33	0.0431	**1,886.30**

The flexural strength of PLA decreased substantially after adding biochar, dropping from 81.33 MPa (pure PLA) to 28.47 MPa and 36.60 MPa for 5% and 10% PLA–biochar composites, respectively. This decrease could be attributed to reductions in interfacial bonding and load-bearing capacity.

Flexural strain increased with biochar content, with 5% PLA–biochar exhibiting a higher strain (0.0827) than pure PLA (0.0431), indicating improved flexibility and greater deformation before failure.

These results highlight a trade-off between strength and flexibility resulting from the incorporation of biochar in PLA composites.

3.3 Wear test

The wear test was conducted using a pin-on-disk tribometer in accordance with the ASTM G99-17 standard to evaluate the tribological behavior of pure PLA, 5% PLA–biochar, and 10% PLA–biochar composite filaments. Wear (mm) and frictional force (N) over time were the key outputs measured, which were subsequently used to derive the coefficient of friction (μ) and wear rate.

Table 3 Wear test results

Sample	Wear Rate (micrometers/second)	Frictional Force (N)	Coefficient of Friction
5% PLA–biochar	0.175	2.627	0.265
10% PLA–biochar	0.125	0.931	0.095
Pure PLA	0.375	0.306	0.375

The wear rate decreased as the biochar content increased, from 0.375 µm/s (pure PLA) to 0.175 µm/s (5%) and 0.125 µm/s (10%), demonstrating improved wear resistance owing to enhanced load-bearing capacity.
The frictional force reduced from 3.06 N (pure PLA) to 2.627 N (5%) and further to 0.931 N (10%), likely due to the lubricating effect of biochar at higher concentrations.
The coefficient of friction exhibited a similar decline, from 0.375 (pure PLA) to 0.265 (5%) and 0.095 (10%), indicating an increase in surface interaction and a decrease in adhesion with increasing biochar content.

3.4 Impact test

The impact test was performed using a Charpy impact tester to determine the behavior of pure PLA, 5% PLA–biochar, and 10% PLA–biochar filaments. The test assessed the impact strength by measuring the impact energy.

Table 4 Impact test results

Sample	Impact Energy (J)	Specimen Width (m)	Impact Strength (J/m)
Pure PLA	2.50	0.01	250
5% PLA–biochar	2.09	0.01	209
10%PLA–biochar	1.96	0.01	196

The impact strength decreased as the biochar content increased, from 250 J/m for pure PLA to 209 J/m (5%) and 196 J/m (10%), indicating a reduction in toughness.
A higher biochar content likely introduces stress concentrators that reduce the material's energy absorption capacity, making it more brittle and prone to fracture under sudden loading.

3.5 Thermal conductivity test

A thermal conductivity test was conducted using the DTC-300 Thermal Conductivity Tester to evaluate the properties of pure PLA, 5% PLA–biochar, and 10% PLA–biochar filaments.

Table 5 Thermal conductivity test results

Temperature (°C)	5% Biochar–PLA (W/mK)	10% Biochar–PLA (W/mK)	Pure PLA (W/mK)
30.0	0.1499	0.1702	0.1372
36.1	0.1566	0.1702	0.1438
42.1	0.1599	0.1702	0.1479
48.2	0.1685	0.1845	0.1533
54.2	0.1822	0.2129	0.1647
60.2	0.1934	0.2414	0.1765

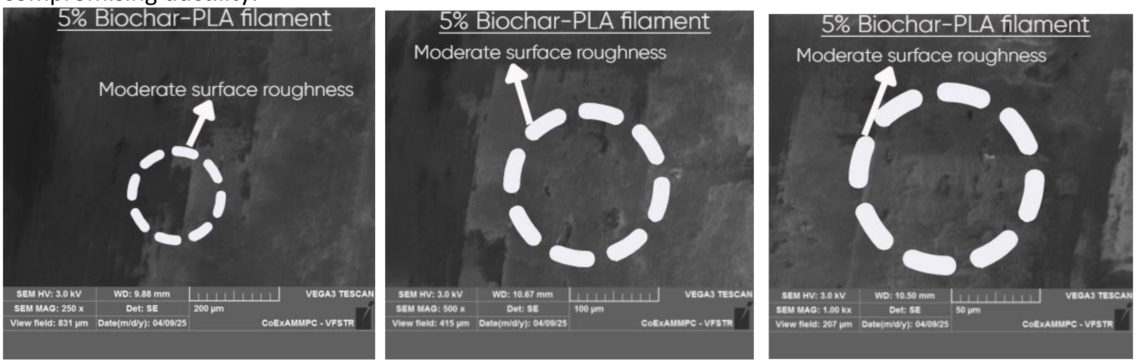
Thermal conductivity increased with the addition of biochar, with 10% PLA–biochar exhibiting the highest values, particularly at elevated temperatures.
This increase was nonlinear, remaining stable at lower temperatures and rising significantly beyond 42°C, signifying improved heat transfer due to enhanced PLA–biochar interactions.

3.6 SEM

SEM was performed on worn-out surface samples with 5 wt.% and 10 wt.% biochar content to understand the wear behavior of PLA–biochar composites. The surface morphology provides insights into the wear mechanisms involved and the effect of the biochar content on material performance.

3.6.1 SEM of 5% PLA–biochar samples

As illustrated in Fig. 23, 24, and 25, the worn-out surface of the 5% PLA–biochar sample exhibited relatively smoother wear tracks, with some visible abrasive grooves and microcracks. The matrix appeared to be well bonded with the filler particles, with few signs of particle pull-out. These features suggest a balanced wear behavior, with the biochar providing reinforcement without considerably compromising ductility.

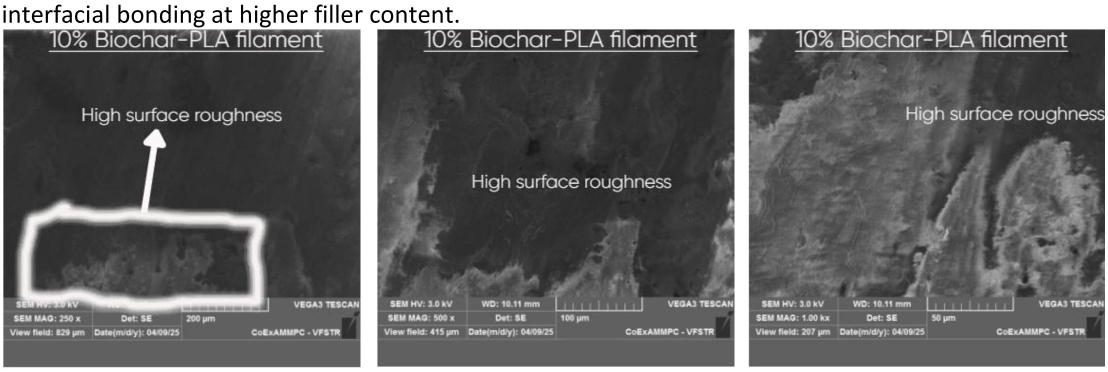


(a) (b) (c)

Fig. 7 5% Biochar–PLA wear test specimen (a) 250x (b) 500x (c) 1000x magnification

3.6.2 SEM of 10% PLA–biochar samples

In contrast, the 10% PLA–biochar samples (Fig. 26, 27, 28) displayed a rougher and more damaged surface. Visible cracks, deep grooves, and clear signs of filler particle pull-out were evident. The wear marks were more irregular, and areas of matrix–filler detachment were noted. These features indicate increased brittleness and reduced wear performance, likely due to biochar agglomeration and poor interfacial bonding at higher filler content.



(a) (b) (c)

Fig. 8 10% Biochar–PLA wear test specimen (a) 250x (b) 500x (c) 1000x magnification

Table 6 Comparison table for SEM images of wear test specimens

Parameter	5% PLA–biochar	10% PLA–biochar
Surface roughness	Moderate	High
Cracks	Few and shallow	Many and deep
Filler pull-out	Rare	Frequent
Matrix–filler bond	Good	Weak/compromised
Wear mechanism	Abrasive wear	Abrasive + brittle fracture
Overall wear resistance	Improved	Reduced

The 5% PLA–biochar composite exhibited smoother worn-out surfaces with fewer shallow cracks, indicating more uniform wear and improved surface integrity.
The filler particles mostly remained embedded in the 5% composite, suggesting strong interfacial bonding and limited filler pull-out.
Abrasive wear dominated in the 5% sample, reflecting effective matrix–filler interaction under sliding contact.
The 10% PLA–biochar composite demonstrated rougher surfaces, deeper cracks, and more frequent filler pull-out, showing poor dispersion and weak bonding.
Although the 10% sample had a lower measured wear rate, SEM revealed signs of brittle fracture and reduced structural integrity, indicating diminished wear resistance due to excessive filler.

3.7 Discussion on nonlinear mechanical behavior

The observed increase in tensile strain at 10 wt.% PLA–biochar, despite weak interfacial bonding, suggests a complex interplay between filler dispersion and matrix deformation. At a low biochar content (5 wt.%), filler particles may restrict the mobility of polymer chains, leading to stiffening and reduced strain. However, as the biochar content increases, partial agglomeration and heterogeneous distribution may create localized stress-relief zones, allowing the material to deform to a greater extent before fracture, thereby increasing the tensile strain. Similar trends have been reported by Yan et al. [34], who found that low biochar loadings improved ductility, whereas higher loadings led to stress-localized deformation.
A comparable trend was observed in flexural and impact tests. Flexural strain increased with the biochar content, and although there was an overall decrease in impact strength, the reduction from 5 wt.% to 10 wt.% biochar was less pronounced than expected. This finding suggests that the filler–matrix interactions occurring at a high biochar content may redistribute the applied stresses nonuniformly, marginally improving ductility or energy absorption in specific regions of the composite. Such nonlinear mechanical responses have been reported in PLA–biochar systems, where low filler contents stiffen the matrix but moderate additions promote microscale stress redistribution, resulting in atypical strain or toughness behavior [33,34].

3.8 Limitations of the study

This study employed single-sample testing for each composite composition due to constraints in fabrication resources and testing costs. While the obtained results provide clear comparative trends between neat PLA and PLA–biochar composites, further studies with larger sample sizes and statistical validation are required to strengthen the findings.

4 Conclusion

This work demonstrates that the addition of 10% biochar significantly enhances the wear resistance and thermal conductivity of PLA, albeit at the expense of compromising its tensile strength. The study confirms the trade-off between mechanical strength and tribological/thermal performance in PLA–biochar composites. This finding offers valuable design insights for engineering applications where such trade-offs are acceptable or desirable, such as in wear-resistant, lightweight components.

- Tensile property analysis revealed that the addition of biochar reduces tensile strength, with 5% causing a sharp drop due to poor bonding, but 10% leading to slight recovery and improved ductility [21].
- Flexural behavior test showed decreased strength but increased flexibility at 5%, with 10% offering better reinforcement and a balance of stiffness and strain [21].
- Wear tests confirmed that biochar enhances wear resistance, with 10% exhibiting the least material loss under sliding contact.
- Frictional behavior improved at a high biochar content, with 10% reducing both the frictional force and the coefficient of friction owing to its lubricating effect.
- Impact strength decreased slightly with the addition of biochar, indicating increased brittleness while maintaining moderate resistance to sudden loads.
- Thermal conductivity improved with the addition of biochar, with 10% biochar–PLA displaying significantly elevated heat dissipation capacity.
- SEM analysis revealed improved wear resistance and bonding in 5% PLA–biochar; in contrast, 10% PLA–biochar exhibited brittle failure due to poor filler dispersion and extensive surface damage.

Biochar reinforcement modifies PLA's performance by reducing its tensile, impact, and flexural strengths but improving its tensile strain, flexural strain, wear resistance, thermal conductivity, and frictional behavior. While 10% PLA–biochar showed the lowest wear rate and friction, ideal for thermal and low-friction applications, SEM analysis revealed that 5% PLA–biochar exhibited better surface integrity and bonding, making it more suitable for durable, load-bearing use.

Future research should focus on optimizing printing parameters to enhance interfacial bonding, evaluating long-term durability under cyclic loading, and exploring hybrid fillers or surface treatments to further improve the performance. In addition, the environmental impact and biodegradability of PLA–biochar composites under real-world conditions warrant detailed investigation.

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